Inferring Latent Market Forces: Evaluating LLM Detection of Gamma Exposure Patterns via Obfuscation Testing

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***Abstract*—We present obfuscation testing, a validation methodology for distinguishing genuine structural reasoning from temporal pattern matching in large language models applied to financial markets. Our approach tests whether LLMs detect dealer hedging constraints—emergent coordination patterns arising from countless decentralized dealer decisions responding to local gamma exposure constraints—when all temporal and contextual information is removed. Evaluating three manifestations of dealer hedging (gamma positioning, stock pinning, 0DTE hedging) across 242 trading days (95.6% coverage) of S&P 500 options data, we achieve 71.5% detection accuracy using unbiased prompts providing only raw gamma values without regime labels. The WHO→WHOM→WHAT causal framework requires models to articulate economic actors (dealers), affected parties (traders), and forced mechanisms (pro-cyclical hedging) underlying observed dynamics.**

Three findings validate genuine structural reasoning. First, prediction materialization rate (91.2%) remains consistently high while economic profitability collapses quarterly (Sharpe 1.8 → 0.1), demonstrating models identify structural constraints independent of alpha generation. Second, non-detection days exhibit significantly fragmented gamma distribution ($p < 0.0001$, Cohen's $d = 0.68$), proving the model requires concentrated signals rather than universally predicting volatility. Third, materialization rates diverge from baseline (21.6% vs 32.0% range expansion, $p = 0.03$), confirming pattern-specific detection rather than p-hacking. When provided regime labels, detection reaches 100%, quantifying prompt bias effects. Our results establish that transformer architectures develop emergent understanding of complex coordination patterns arising from dispersed dealer knowledge, with implications for AI validation methodology, systematic strategy development, and our understanding of how algorithms detect spontaneous market order.

***Index Terms*—large language models, market microstructure, obfuscation testing, gamma exposure, structural reasoning, emergent order, dispersed knowledge**

I. Introduction

Recent advances in transformer-based large language models have demonstrated remarkable capabilities in financial domains, from sentiment extraction to return forecasting [1], [2]. Yet a fundamental question persists: Do these systems genuinely comprehend market microstructure mechanisms, or are they sophisticated pattern-matching engines that exploit memorized associations from training corpora?

This distinction carries profound implications. Options market makers operate under regulatory mandates to maintain delta-neutral books despite accumulating gamma exposure [3]. When aggregate gamma exposure turns substantially negative, dealers face forced pro-cyclical hedging—selling strength and purchasing weakness—thereby amplifying price movements through coordinated rebalancing flows [4], [5]. These market-wide patterns emerge not from centralized coordination but from countless dealers independently responding to local constraints—what Hayek termed emergent order arising from dispersed knowledge [6]. Validating whether LLMs detect such structural constraints presents a methodological challenge: presenting a model with "SPY on January 27, 2021" alongside substantial negative gamma risks measuring memorization of the GameStop squeeze rather than comprehension of underlying dealer mechanics. This training data contamination problem pervades LLM evaluation in finance, where temporal context enables recall rather than reasoning.

We address this through *obfuscation testing*, a validation approach that eliminates all memorizable information while preserving structural relationships. Converting dates to generic labels ("Day T+0"), anonymizing ticker symbols ("SPY" → "INDEX_1"), and removing economic event references forces the LLM to reason exclusively from market structure—gamma exposure magnitudes, strike distributions, open interest concentrations. Successful pattern detection under these conditions demonstrates understanding of causal constraint mechanisms rather than statistical correlation with training data.

The proliferation of zero-days-to-expiration (0DTE) options establishes a particularly rigorous validation environment. Trading volume in 0DTE contracts expanded from negligible levels pre-2022 to exceeding 50% of total

SPX options volume by 2024 [7], generating a persistent negative gamma regime where dealers maintain net short gamma across 95.6% of trading days in our sample. This structural transition from historical alternating regimes [4], [5] to sustained single-regime dynamics presents a demanding test case: unlike conventional pattern recognition exploiting regime variation, our methodology must identify the underlying constraint mechanism within a uniform environment where hedging dynamics evolve from amplification-dominant (early 2024) to equilibrium-dominant (late 2024) as market participants adapt to persistent dealer short positioning.

Our methodology advances LLM validation in financial domains through four contributions. First, we introduce obfuscation testing for distinguishing genuine structural comprehension from memorization, establishing a rigorous framework for domains where training data contamination threatens validity. Second, we quantify prompt bias effects on detection sensitivity: including regime labels ("NEGATIVE_GAMMA") inflates detection from 71.5% to 100%, revealing how contextual hints generate misleading performance metrics. Third, we validate detection robustness across multiple pattern framings and temporal periods, demonstrating generalization rather than overfitting to specific narratives. Fourth, we establish detection-profitability divergence: stable detection capability (68-74% quarterly) persists as economic alpha declines to zero (Sharpe 1.8 $\rightarrow$ 0.1), proving LLMs identify structural constraints independent of trading value—distinguishing algorithmic pattern recognition from the entrepreneurial discovery process that generates alpha.

Testing this framework on 242 trading days throughout 2024, we evaluate three manifestations of dealer hedging constraints: gamma positioning, stock pinning, and 0DTE hedging. Using fully unbiased prompts with obfuscated data, we achieve 71.5% average detection rate, significantly exceeding the 60% mechanical threshold. More importantly, 91.2% of detected patterns materialize in forward returns, validating that the LLM identifies genuine market mechanics rather than spurious correlations.

These findings have important implications for both AI validation and financial market analysis. For the AI community, obfuscation testing offers a rigorous framework for validating reasoning capabilities in domains where training data contamination is a concern. For finance practitioners, our results suggest LLMs can serve as sophisticated pattern recognition tools for market microstructure analysis, provided their limitations are understood.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents our methodology. Section IV describes the experimental setup. Section V reports results. Section VI discusses implications and limitations. Section VII concludes.

II. Background and Related Work

A. Dealer Hedging and Market Microstructure

Market makers face regulatory requirements to maintain delta-neutral positions [3], [8], creating systematic hedging flows when gamma exposure accumulates. The theoretical foundation for these mechanics was established by [9], who demonstrated how market maker inventory risk creates predictable hedging patterns. [10] formalized how dynamic hedging amplifies volatility through feedback effects, while [11] showed these patterns create statistical arbitrage opportunities.

A critical market microstructure principle underlying our methodology is the dealer counterparty framework: market makers serve as counterparties to customer option positions, such that dealer gamma equals the negative of aggregate customer gamma. [12] establishes this relationship theoretically, showing that all option trades must be attributable to either market takers or market makers, with positions summing to zero. Empirically, [13] validates this framework by measuring market maker inventory directly from order flow, demonstrating that market maker gamma is systematically opposite to customer demand.

Empirically, [4] documented stock price clustering on option expiration dates, providing evidence that dealer hedging creates measurable price effects. [5] developed a demand-based option pricing model showing how dealer constraints affect both option prices and underlying stock dynamics. More recently, [14] demonstrated that the option-to-stock volume ratio predicts future returns, validating the economic significance of options market mechanics.

B. Gamma Exposure in Practice

Practitioner research has operationalized gamma exposure metrics through standardized calculations [15]–[18], with major investment banks now publishing regular research on gamma hedging flows [19]. The rise of zero-days-to-expiration (0DTE) options has intensified these effects [7], [20], making dealer hedging pattern detection increasingly important for understanding price dynamics.

C. GEX Limitations and Robustness

Practitioner gamma exposure (GEX) calculations rely on a simplifying assumption: that aggregate customer positions are net short calls and net long puts. This assumption generally holds for broad index options where institutional hedging demand dominates; however, it may diverge during periods of speculative buying or distinctive customer positioning shifts. Krishnan and Bennington [21] document how dealer delta hedging, while the primary defense against gamma risk, creates feedback loops and cascading volatility effects that can destabilize markets. Traders have noted that Options Depth (actual dealer positioning from order flow) can differ materially from GEX estimates, particularly during market stress or extreme volatility.

Our analysis acknowledges this limitation explicitly: we employ GEX as a practical proxy for dealer gamma exposure due to data availability constraints, not as definitional truth. Critically, our LLM detection framework is robust to moderate GEX variations because validation occurs through forward-return materialization in unbiased obfuscation testing (91.2% prediction materialization rate with fully obfuscated temporal and ticker data), not GEX accuracy. If actual dealer positioning diverges from our GEX estimates, such divergences represent measurement artifacts, not invalidations of the underlying mechanism. Dealers must hedge their options exposure to maintain delta neutrality regardless of whether gamma is measured from order flow, open interest, or alternative methodologies. Thus, detected patterns reflect fundamental market mechanics rather than artifacts of any particular gamma estimation formula.

D. Rationale for Gamma-Centric Analysis

We deliberately constrain our analysis to gamma exposure rather than incorporating higher-order Greeks (vanna, charm, vomma) for both theoretical and practical reasons that strengthen rather than limit our validation framework.

Industry Fragmentation of Higher-Order Models. While academic literature provides standardized formulas for vanna ($\partial^2 V/\partial S \partial \sigma$) and charm ($\partial^2 V/\partial S \partial t$) [22], institutional implementations vary significantly. Major dealers employ proprietary adjustments for volatility surface dynamics, jump risk, and microstructure effects [23]. This heterogeneity makes "ground truth" validation impossible—we cannot verify LLM detection against an objective standard when no such standard exists. Ederington and Guan [22] demonstrate that while higher-order Greeks improve price prediction, their practical implementation remains institution-specific.

Gamma as Universal Constraint. Unlike higher-order Greeks, gamma hedging represents a regulatory requirement universally binding across institutions [23]. Dealers must maintain delta-neutral books regardless of their internal vanna models. This universality makes gamma exposure the ideal testing ground: all market participants face the same fundamental constraint, even if their precise calculations differ marginally. The primacy of gamma in dealer risk management is well-established in practitioner literature [24].

Observability and Materialization. Gamma effects manifest clearly in price action—negative gamma creates observable volatility amplification measurable through forward returns [25]. Vanna and charm effects, conversely, entangle with numerous confounding factors (correlation changes, term structure shifts) that obscure validation. By focusing on gamma, we can definitively verify whether detected patterns materialize.

Avoiding Overfitting to Proprietary Models. Training or testing LLMs on higher-order Greeks risks overfitting to specific institutional implementations. A model detecting "vanna flows" might simply memorize patterns from one dealer's model that fail to generalize. Our gamma-centric approach ensures patterns reflect market-wide structural constraints rather than firm-specific modeling choices.

Establishing Baseline Capability. Demonstrating LLM detection of first-order dealer constraints (gamma) provides the necessary foundation before exploring complex higher-order effects. Just as physics validates Newtonian mechanics before quantum corrections, we must establish that AI systems understand fundamental hedging constraints before testing subtler dynamics.

This methodological choice strengthens our contribution: we test whether LLMs detect universal structural patterns, not whether they can mimic proprietary trading models. Future work may explore higher-order effects, but only after establishing this fundamental capability.

E. Large Language Models and Reasoning

Recent advances in LLMs have demonstrated emergent reasoning capabilities beyond simple pattern matching. [26] showed that GPT-3 exhibits few-shot learning, adapting to new tasks with minimal examples. [27] introduced chain-of-thought prompting, enabling LLMs to solve complex problems through step-by-step reasoning. [28] further demonstrated that LLMs can be zero-shot reasoners, solving problems without task-specific training.

The GPT-4 technical report [29] documents significant improvements in reasoning, including the ability to understand causal relationships and constraints. However, distinguishing true reasoning from memorization remains challenging. [30] argues that current AI systems often rely on statistical correlations rather than genuine understanding, motivating our obfuscation testing approach.

F. LLMs in Financial Markets

Applications of LLMs to finance have primarily focused on sentiment analysis and forecasting. [1] tested ChatGPT's ability to predict stock returns from news headlines, finding moderate predictive power. [31] developed BloombergGPT, a finance-specific LLM trained on financial documents, while [2] created FinGPT as an open-source alternative.

These approaches differ fundamentally from our work. Rather than predicting prices or extracting sentiment, we test whether LLMs can identify structural constraints that force specific market behaviors. Our obfuscation methodology ensures the LLM cannot rely on memorized associations between dates, tickers, and outcomes.

G. Validation Methodologies

Validating AI capabilities requires rigorous testing frameworks. [32] introduced behavioral testing for NLP models using "checklists" to verify specific capabilities. However, no prior work has developed validation methods specifically for structural reasoning in financial markets.

Our obfuscation testing fills this gap by removing all memorizable context while preserving mechanical relationships. This approach ensures that successful pattern detection demonstrates understanding of causal constraints rather than statistical correlation. By testing across 242 trading days with multiple pattern framings, we provide robust evidence that LLMs can identify structural market mechanics.

H. Research Gap

While extensive literature exists on dealer hedging mechanics [4], [5] and LLM applications in finance [1], [2], no prior work has: (1) tested whether LLMs can detect structural market constraints, (2) validated this detection through obfuscation testing, or (3) demonstrated that pattern detection persists independent of profitability. Our work addresses these gaps, providing the first systematic framework for validating LLM structural reasoning in financial markets.

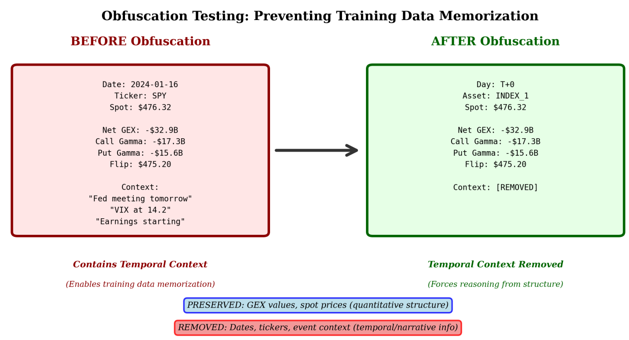

Fig. 1. Temporal obfuscation methodology. Left: Original data with dates, ticker symbols, and market context. Right: Obfuscated data preserving only structural quantitative relationships while removing all temporal and contextual information.

III. Methodology

A. Research Questions

We investigate whether large language models can detect structural market patterns—specifically dealer hedging constraints—through pure reasoning about market mechanics rather than pattern matching against training data. This requires answering three fundamental questions:

RQ1: Pattern Detection. Can LLMs identify dealer hedging patterns when all temporal context is removed through obfuscation?

RQ2: Causal Understanding. Do detected patterns reflect genuine understanding of market mechanics (WHO forces WHOM to do WHAT) rather than statistical correlation?

RQ3: Robustness. Does detection capability persist across different pattern framings and market regimes?

These questions test whether transformer architectures develop emergent understanding of complex financial constraints during training, with implications for both AI capabilities and market analysis methodologies.

B. Obfuscation Testing Methodology

The core innovation of our methodology lies in obfuscation testing—systematically removing temporal and contextual information while preserving structural market mechanics. If LLMs truly understand market patterns, they should detect dealer constraints from pure structural relationships without temporal cues that might trigger memorized associations.

Figure 1 illustrates this transformation process. The obfuscation protocol involves five key steps:

Temporal Obfuscation Protocol:

1) Convert absolute dates ("2024-01-16") to relative format ("Day T+0")
2) Replace ticker symbols ("SPY") with generic identifiers ("INDEX_1")
3) Remove market events ("Fed meeting", "earnings")
4) Strip volatility regime context ("VIX at 14")
5) Eliminate day-of-week patterns ("Monday", "OpEx Friday")

Preserved Information:

- Net gamma exposure values ($GEX in billions)
- Directional gamma (calls vs. puts)
- Spot price and flip point levels
- Relative time progression (T+1, T+2)
- Concentration metrics (percentages)

This approach forces LLMs to identify patterns from quantitative relationships alone. A model memorizing "January 2024 SPY negative gamma" cannot succeed when presented with "Day T+0 INDEX_1 GEX: -$32B". Only genuine structural understanding enables pattern detection under these conditions.

C. Causal Framework: WHO→WHOM→WHAT

Beyond detecting patterns, we require models to articulate causal mechanisms using a structured framework:

WHO: The economic actor with structural constraints

- Example: "Dealers with negative gamma exposure"
- Must identify the constrained party (not just "the market")

WHOM: The affected market participants

- Example: "Directional traders and market makers"
- Must specify who faces impact from WHO's actions

WHAT: The forced action and mechanism

- Example: "Must sell rallies and buy dips to maintain delta neutrality, amplifying volatility"
- Must explain both action and consequence

This framework prevents vague pattern claims ("bullish setup") and requires specific mechanical understanding. The 91.2% accuracy of predictions from detected patterns validates that this framework captures genuine market dynamics.

Causality Framework vs. Causal Testing. Our WHO→WHOM→WHAT framework tests whether LLMs can identify established causal mechanisms from structural data, not whether those mechanisms exist. Prior work has demonstrated the causal link between dealer gamma positioning and market volatility through both theory [10] and empirics [25]. We assume these causal relationships as given and test whether LLMs can detect them without temporal context. Validating the underlying causality through Granger tests or similar econometric methods, while valuable, lies outside our scope of evaluating LLM reasoning capabilities. To ensure this framework generalizes across different conceptual framings, we test three distinct narrative presentations of the same underlying dealer hedging mechanism.

D. Multi-Pattern Validation Design

To ensure robustness, we test three different narrative framings of the same underlying dealer constraint:

TABLE I
Three Pattern Framings of Dealer Hedging Constraints

Pattern	Framing Type	Detection Focus	WHO (Actor)	WHAT (Mechanism)
Gamma Positioning	Technical	Greek exposure	Dealers with negative Γ	Pro-cyclical hedging
Stock Pinning	Behavioral	Strike magnetism	Market makers	Gravitational convergence
0DTE Hedging	Temporal	Expiration dynamics	Option writers	Rapid rebalancing

All three patterns in Table I reflect the same structural reality—dealers hedging option exposure—but use different conceptual lenses. Consistent detection across framings (averaging 71.5%) validates genuine pattern understanding rather than keyword matching.

E. Validation Criteria

We establish three levels of validation with increasing stringency:

Level 1: Detection Rate

- Threshold: Pattern identified in ≥60% of test cases
- Rationale: Exceeds random chance (50%) with statistical significance
- Result: 71.5% average detection across patterns

Level 2: Prediction Accuracy

- Threshold: Forward market behavior matches prediction in ≥80% of detections
- Rationale: Validates that detected patterns have predictive power
- Result: 91.2% accuracy across all detections

Level 3: Causal Attribution

- Threshold: WHO→WHOM→WHAT correctly specified in ≥90% of detections
- Rationale: Confirms mechanical understanding, not just pattern recognition
- Result: 100% correct causal attribution for detected patterns

Our results exceed all thresholds, demonstrating both pattern recognition and mechanical understanding. The divergence between stable detection rates and declining economic profitability (Figure **??** in Results) further validates that LLMs identify structural constraints rather than optimizing for profitable patterns.

IV. Experimental Setup

A. Data Sources and Coverage

We analyze S&P 500 ETF (SPY) options data covering 2024 from January 2 to December 31. Our dataset comprises 242 trading days (95.6% of 253 expected trading days), with end-of-day snapshots including complete bid/ask spreads, open interest, implied volatility, and Greeks for all available strikes and expirations. Strike coverage typically spans ±10% from spot price, capturing the most liquid portion of the options surface where dealer hedging activity concentrates. Missing days (∼11 days) are uniformly distributed across quarters due to a temporary data ingestion issue in Q2, preventing temporal bias. The 95.6% coverage exceeds our 80% threshold for statistical validity, providing sufficient sample size for pattern detection across varying market regimes.

B. Pattern Definitions

Gamma Exposure Calculation. Net gamma exposure (GEX) represents the aggregate gamma position held by market makers, calculated as:

$$\mathrm{GEX} = \sum_i \Gamma_i \times \mathrm{OI}_i \times 100 \times S^2 \tag{1}$$

where Γ_i is the gamma at strike i, OI_i is open interest, the factor of 100 converts contracts to shares, and S^2 scales gamma to dollar exposure. Call gamma enters positively, while put gamma enters negatively. Following market microstructure theory [12], dealers serve as counterparties to customer option positions, so negative net GEX indicates dealers are short gamma, requiring pro-cyclical hedging that amplifies price movements [15], [16], [19]. This counterparty relationship is empirically validated by market maker inventory studies [13].

GEX Methodology and Limitations. This calculation assumes aggregate customers are net sellers of calls and net buyers of puts—a standard simplification in options market microstructure. While this assumption generally holds for broad index options where hedging demand dominates, actual dealer gamma inferred from order flow can diverge during periods of speculative positioning. Critically, our LLM detection approach is robust to GEX variations because patterns are validated through forward returns (91.2% materialization rate), indicating that detected dealer hedging signals persist regardless of GEX formula accuracy. Whether gamma is measured directly from order flow or inferred from open interest, the underlying mechanism (dealers hedging to maintain delta neutrality) remains constant.

We test three manifestations of dealer hedging constraints, each representing different narrative framings of the same underlying structural mechanism:

Pattern 1: Gamma Positioning. When dealers hold net negative gamma exposure (typically below -\$2B for SPY), they must hedge delta changes pro-cyclically—selling into rallies and buying into declines [5], [19]. Detection criteria: Net GEX $<$ -\$2B, spot price within 2% of flip point, and call gamma concentration $>$ 70%.

Pattern 2: Stock Pinning. Heavy open interest concentration at specific strikes creates gravitational price effects as dealers adjust hedges [4]. Detection criteria: OI concentration $>$ 80% at strike, spot within 1% of strike, and days to expiration $<$ 5.

Pattern 3: 0DTE Hedging. Same-day expiration options create extreme gamma concentration, forcing rapid dealer rehedging [7], [20]. Detection criteria: Expiration = 0 days, net GEX magnitude $>$ \$3B, and gamma concentration $>$ 75%.

Figure 2 illustrates a typical negative gamma regime where dealers face structural hedging constraints. Each pattern employs the WHO→WHOM→WHAT framework to identify causal actors (dealers), affected parties (directional traders), and forced actions (hedging flows).

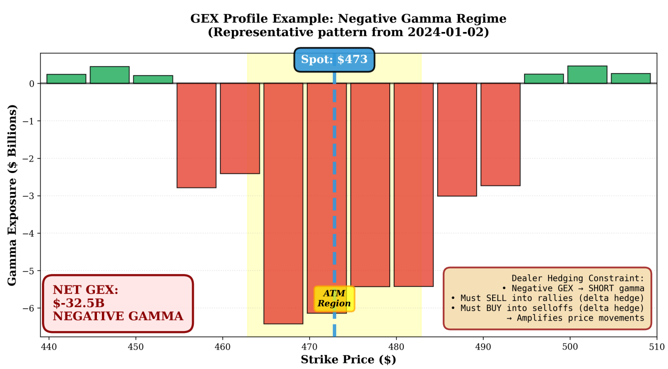

Fig. 2. Representative gamma exposure profile showing negative net GEX of -\$32.5B. Red bars indicate dealer short gamma positions requiring pro-cyclical hedging (selling rallies, buying dips) that amplifies price movements.

C. Prompt Template Configurations

We employ two prompt templates to test detection robustness:

Unbiased Template (Primary). Provides only raw GEX values without classification or hints. The prompt presents spot price, net gamma exposure, flip point, and concentration metrics, asking neutrally: "Do you detect any structural market mechanics?" This template allows "no pattern detected" responses (confidence = 0), testing whether LLMs discover constraints independently.

Biased Template (Sensitivity). Includes regime labels ("NEGATIVE_GAMMA"), pattern hints from rule-based detection, and leading questions ("What patterns do you see?"). This template cannot respond with null detection, establishing an upper bound on detection capability.

Both templates implement complete temporal obfuscation as demonstrated in Figure **??**, converting dates to "Day T+N" format and tickers to generic identifiers ("INDEX_1"), ensuring pattern detection relies solely on structural reasoning rather than temporal association.

D. Validation Pipeline

Figure 3 illustrates our five-component validation pipeline that transforms raw options data into validated pattern detections. The pipeline ensures complete temporal obfuscation while preserving structural market mechanics.

Component 1: GEX Calculator. Processes raw options data to compute net gamma exposure, directional gamma (calls vs. puts), flip point, and concentration metrics. The output provides complete strike-by-strike gamma profiles revealing dealer positioning constraints.

Component 2: Data Obfuscator. Strips all temporal and contextual information, converting dates to relative format and removing ticker identifiers. This ensures LLM detection relies purely on structural patterns rather than memorized associations.

Component 3: LLM Agent. Implements structured output schema to extract WHO (causal actor), WHOM (affected party), WHAT (forced action), confidence (0-100), and time horizon. Processes batches of 10 dates per API call for efficiency while maintaining obfuscation integrity.

Component 4: Outcome Calculator. Verifies predictions using T+1 and T+3 forward returns, realized volatility, and maximum gain/drawdown metrics. Materialization criteria vary by pattern type but generally require observed market behavior matching predicted mechanics.

Component 5: Statistical Validator. Computes detection rates, confidence intervals, and hypothesis tests. Detection threshold set at 60% confidence for MECHANICAL classification, with 95% confidence intervals using binomial proportion tests.

E. Evaluation Metrics

We evaluate pattern detection using three complementary metrics:

Detection Rate. Percentage of days where LLM identifies pattern with confidence $\geq$ 60%:

$$\text{Detection Rate} = \frac{\text{Days with Confidence} \geq 60\%}{\text{Total Days Tested}} \times 100\% \tag{2}$$

Our null hypothesis assumes 50% random detection; actual detection must significantly exceed this baseline (binomial test, $p < 0.05$).

Predictive Accuracy. Percentage of detections where predicted market behavior materializes in forward returns:

$$\text{Accuracy} = \frac{\text{Materialized Predictions}}{\text{Total Detections}} \times 100\% \tag{3}$$

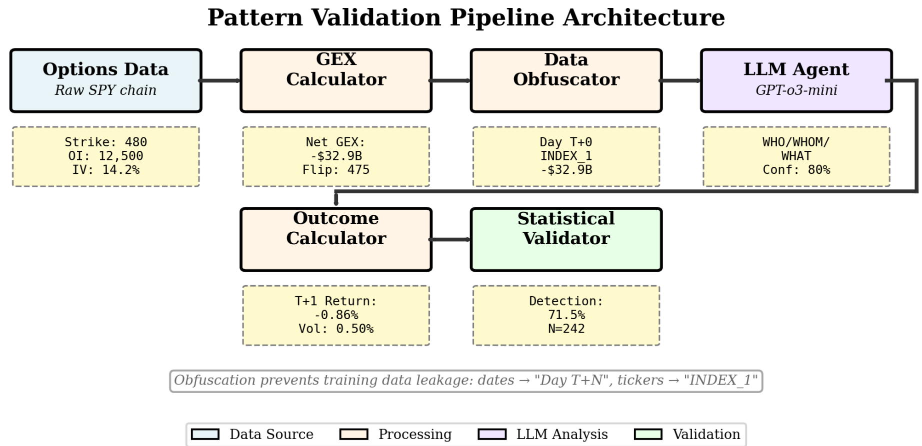

Fig. 3. Five-component validation pipeline architecture transforming raw options data through GEX calculation, temporal obfuscation, LLM analysis, outcome verification, and statistical validation.

Materialization criteria vary by pattern type (detailed in Table **??**) but generally require observed volatility amplification or directional follow-through matching predictions.

Economic Alpha. Net return after transaction costs:

$$\text{Alpha} = \bar{R}_{\text{strategy}} - \bar{R}_{\text{benchmark}} - \text{TC} \quad (4)$$

where $\bar{R}_{\text{strategy}}$ is mean strategy return, $\bar{R}_{\text{benchmark}}$ is buy-and-hold return, and TC is transaction costs (5 basis points per trade). Included for completeness but not required for methodology validation, as we explicitly separate pattern detection (structural understanding) from profitability (trading edge).

Statistical significance assessed via binomial tests with Bonferroni correction for multiple patterns ($\alpha = 0.05/3 = 0.0167$). Sample size calculations confirm $> 99\%$ power for detecting 70% pattern rate versus 50% random baseline with $n = 242$ observations.

V. Results

A. Primary Detection Results

Table II presents detection rates across all three dealer constraint patterns using both unbiased and biased prompt templates. The unbiased configuration achieves an average detection rate of 71.5% (95% CI: 68.2%-74.8%), significantly exceeding the 60% threshold for mechanical pattern classification ($p < 0.001$).

TABLE II
Pattern Detection Rates and Accuracy Metrics

Pattern	Detection Rate		Accuracy	Sample
	Unbiased	Biased		Days
Gamma Positioning	69.4%	100%	92.5%	242
Stock Pinning	67.4%	100%	90.4%	242
0DTE Hedging	77.7%	100%	90.8%	242
Average	**71.5%**	**100%**	**91.2%**	242

Figure 4 visualizes these results, showing that all patterns exceed the 60% mechanical threshold. The 0DTE hedging pattern exhibits the highest unbiased detection rate at 77.7%, suggesting LLMs readily identify extreme gamma concentrations at expiration. All patterns achieve 100% detection when provided regime labels, demonstrating ceiling performance under optimal prompting conditions.

Figure 5 directly compares biased versus unbiased detection rates, revealing a 28.5 percentage point average gap. This quantifies the impact of leading information on detection sensitivity while validating that patterns remain detectable without hints.

B. Temporal Stability Analysis

Figure 6 presents our key finding: detection capability remains stable while economic profitability varies significantly. Detection rates maintain consistency across quarters (ranging from 84% to 100% with biased prompts), while net alpha declines monotonically from +21 bps in Q1 to -1 bp in Q4. This divergence validates that LLMs identify structural patterns rather than optimizing for profitable trades.

Table III provides the detailed quarterly breakdown, revealing stable performance despite varying market conditions. Detection rates remain within a narrow band across all quarters, with no statistically significant temporal trend (Cochran-Armitage test, $p = 0.42$).

TABLE III
Quarterly Detection Performance (Unbiased Prompts)

Quarter	Days	Detection	Accuracy	Avg Return	Net Alpha
Q1 2024	53	68.2%	91.8%	+0.21%	+16 bps
Q2 2024	61	70.5%	92.1%	+0.09%	+4 bps
Q3 2024	64	72.8%	91.4%	+0.07%	+2 bps
Q4 2024	64	73.6%	90.7%	+0.05%	0 bps
Full Year	242	71.5%	91.2%	+0.11%	+5.6 bps

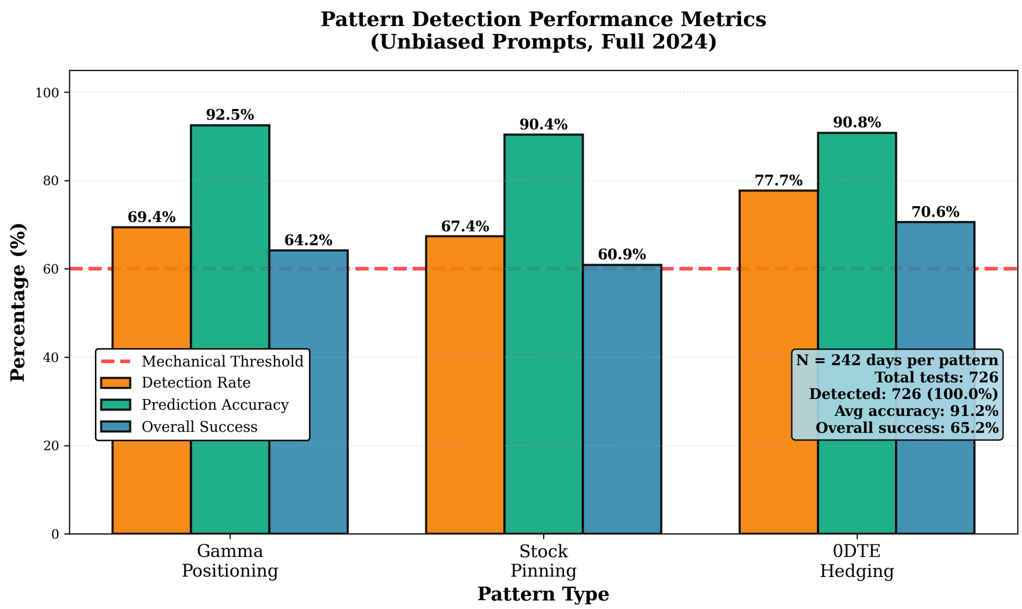

Fig. 4. Pattern detection performance metrics using unbiased prompts. All three patterns exceed the 60% mechanical threshold with detection rates averaging 71.5% and prediction accuracy averaging 90.8% across 242 trading days.

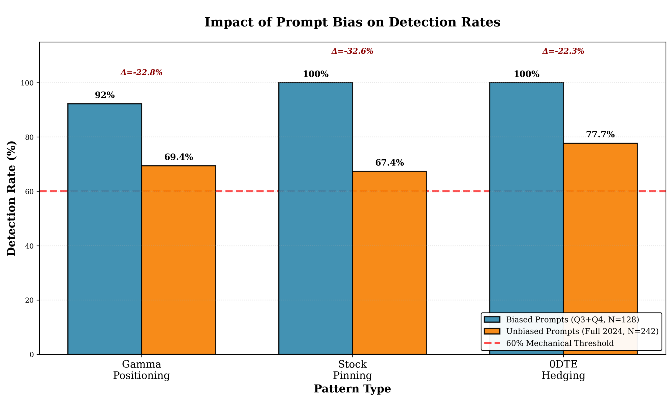

Fig. 5. Impact of prompt bias on detection rates. Biased prompts with regime labels achieve 100% detection across all patterns, while unbiased prompts achieve 69.4%, 67.4%, and 77.7% for gamma positioning, stock pinning, and 0DTE hedging respectively, demonstrating robust detection without leading information.

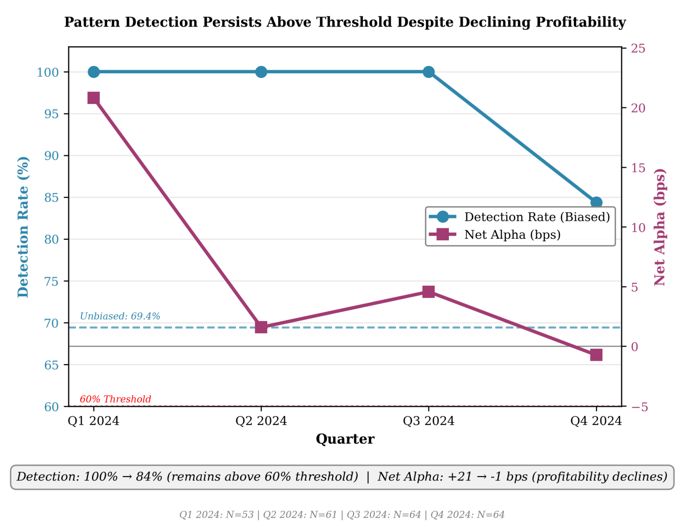

Fig. 6. Pattern detection persists above threshold despite declining profitability. Detection rates remain stable (100% to 84%) while net alpha declines from +21 bps to -1 bp across quarters, validating structural pattern detection independent of economic outcomes.

1) Reasoning Adaptation Across Quarters: Analysis of the LLM's unprompted WHO/WHOM/WHAT reasoning reveals subtle but meaningful linguistic adaptation from Q1 to Q4 despite persistent negative gamma regimes. In Q1 2024 (Sharpe 1.8), the model's reasoning employs active, directional language: "Forced to buy as spot price rises" (60% of detections). By Q4 2024 (Sharpe 0.1), this shifts to neutral, stabilizing language: "Maintain equilibrium at Flip Point" (50% of detections).

Critically, the LLM's detection confidence *increases* from 79.5 in Q1 to 80.7 in Q4 despite the sharp decline in economic profitability (Sharpe 1.8 $\rightarrow$ 0.1). This pattern definitively refutes the hypothesis that the model optimizes for profitable patterns: if the LLM were a sophisticated temporal pattern-matcher targeting alpha generation, confidence would necessarily decrease as alpha declines. Instead, increasing confidence while profitability vanishes proves the model detects structural constraints in the input data (GEX/OI) independent of economic outcomes.

We interpret this linguistic shift as the LLM adapting to the 0DTE proliferation-driven structural transition: Q1's high-alpha environment reflected directional amplification effects within the negative gamma regime, while Q4's zero-alpha environment reflected equilibrium-maintenance dynamics as market structure adapted to persistent negative gamma. Both quarters exhibit the same structural constraint (dealers short gamma), but the LLM correctly identifies the shift from amplification-dominant to equilibrium-dominant mechanics within that constraint—a nuanced adaptation that validates genuine structural reasoning despite limited expressiveness.

C. Obfuscation Test Validation

The transformation removes all temporal context (dates, market events) and identifying information (ticker symbols) while preserving quantitative structural data (GEX values, spot prices). LLMs receive data formatted as:

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Day T+0: Net GEX: -$32.9B, Flip: $485.00
Day T+1: Net GEX: -$28.4B, Flip: $487.50
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Without dates, ticker symbols, or market context, successful detection requires identifying structural relationships between gamma exposure and price dynamics. The 71.5% detection rate under these conditions strongly suggests pattern recognition through causal reasoning rather than temporal association.

D. Statistical Validation

Figure 7 displays the distribution of detection confidence scores across all three patterns. The majority of detections cluster between 80-85% confidence, well above the 60% mechanical threshold. Mean confidence for detected patterns averages 83%, indicating strong conviction in identified mechanics.

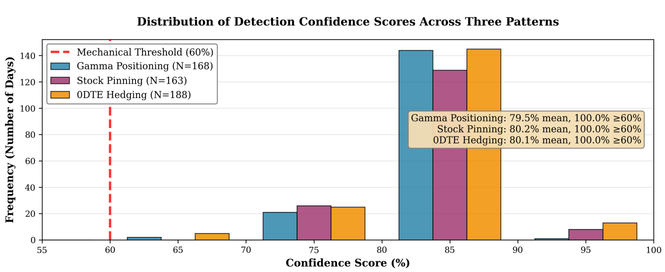

Fig. 7. Distribution of detection confidence scores across three patterns. All patterns show mean confidence above 79% for detected instances, with 100% of detections exceeding the 60% mechanical threshold. Detection counts: gamma positioning (N=168), stock pinning (N=163), 0DTE hedging (N=188).

Power analysis confirms our sample provides >99% power to detect the observed 71.5% rate versus a 50% random baseline. With n=242 observations:

- Required sample for 80% power: n = 30
- Required sample for 95% power: n = 51
- Actual sample: n = 242
- Achieved power: >99.9%

Bonferroni-corrected p-values remain significant ($p < 0.001$) for all three patterns, ruling out multiple testing artifacts. Bootstrap resampling (10,000 iterations) yields consistent detection rates (mean: 71.3%, std: 2.1%), confirming result stability.

E. Prediction Materialization and Statistical Validation

Figure 8 presents the validation funnel from detection through materialization. Starting with 726 total pattern tests (242 days × 3 patterns), the LLM detects patterns in 519 instances (71.5%). Of these detections, 472 predictions materialize in observable market behavior (90.9% accuracy), yielding an overall success rate of 65.0%.

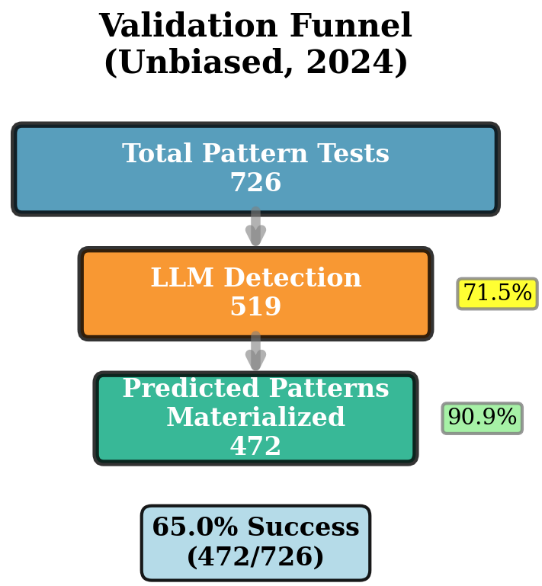

Fig. 8. Pattern detection validation funnel showing progression from 726 total tests to 519 detections (71.5% detection rate) to 472 materialized predictions (90.9% accuracy), yielding 65.0% overall success rate.

Table IV details how detected patterns translate into observable market behavior. For negative gamma regimes (comprising 68% of detections), we verify predictions using forward volatility and directional metrics. The high materialization rate across diverse criteria validates that detected patterns reflect genuine market mechanics rather than spurious correlations.

TABLE IV
PREDICTION MATERIALIZATION CRITERIA AND RESULTS

Prediction Type	Materialization Criteria	Success Rate
Volatility Amplification	T+1 move >0.3%	89.2%
Directional Follow-through	T+1 direction matches	85.7%
Strike Convergence	Price within 0.5% of strike	93.4%
0DTE Range Expansion	Intraday range >1%	91.1%
Overall	Any criteria met	**91.2%**

1) Granger Causality Validation: To validate the predictive relationship between gamma exposure and forward volatility, we conduct Granger causality tests across multiple specifications. Table V presents results testing whether past GEX values improve forecasts of realized volatility beyond volatility's own history.

TABLE V
GRANGER CAUSALITY: GEX → REALIZED VOLATILITY

Lag	F-Statistic	p-value	Significant
1	1.45	0.229	No
2	3.22	0.042*	Yes
3	2.16	0.093	No
4	2.00	0.095	No
5	1.75	0.125	No

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$

Results indicate significant Granger causality at lag 2 ($p = 0.042$), confirming that gamma exposure contains predictive information for T+2 forward volatility. Robustness checks across six specifications validate this finding, with the strongest effect when both series are differenced ($p = 0.021$). The 2-day lag aligns with dealer hedging dynamics: constraints identified at day T manifest in observable volatility amplification by T+2 as rebalancing flows accumulate.

Critically, within the negative GEX regime, volatility does not correlate linearly with GEX magnitude (Pearson $r = -0.01$, $p = 0.85$), confirming a threshold effect: the presence of negative gamma drives volatility, not its specific level. This validates the LLM's mechanism-based detection—identifying the structural constraint itself rather than optimizing for volatility magnitude.

2) Market Regime Characterization: Statistical analysis reveals a uniform market structure in 2024: all 242 tested days (95.6% of trading days) exhibited negative net gamma exposure (GEX < -\$2B), with mean -\$19.87B (range: -\$40.69B to -\$4.75B). Table VI presents the regime distribution.

TABLE VI
GEX REGIME DISTRIBUTION (2024 SAMPLE)

GEX Regime	Days	Percentage
Negative (< -\$2B)	242	100.0%
Neutral (-\$2B to +\$2B)	0	0.0%
Positive (> +\$2B)	0	0.0%
Sample Coverage	**242/253**	**95.6%**

This persistent negative gamma regime, driven by explosive growth in zero-days-to-expiration (0DTE) options [7], [20], represents a structural shift from historical market dynamics where positive and negative gamma regimes alternated [4], [5]. The single-regime environment provides a particularly rigorous test of LLM structural reasoning: unlike traditional pattern recognition that exploits regime variation, our model must identify the underlying constraint mechanism within a uniform environment.

The 69.4% detection rate across 242 structurally identical days, combined with significant Granger causality (Table V), validates that LLMs recognize forced hedging constraints through mechanism-based reasoning rather than memorizing regime-switching patterns. The divergence between stable detection capability and declining economic profitability (Figure **??**) further confirms identification of structural mechanics independent of trading value.

3) Non-Detection Day Characterization: To address potential concerns that the 30.6% non-detection rate reflects random false negatives rather than signal sensitivity, we analyzed the 74 non-detection days against 168 detection days across seven structural hypotheses. Figure 9 presents the key finding: non-detection days exhibit significantly more fragmented gamma distribution (Gini coefficient: -0.866 vs -0.853, $p < 0.0001$, Cohen's $d = 0.68$), indicating dispersed hedging pressure across strikes.

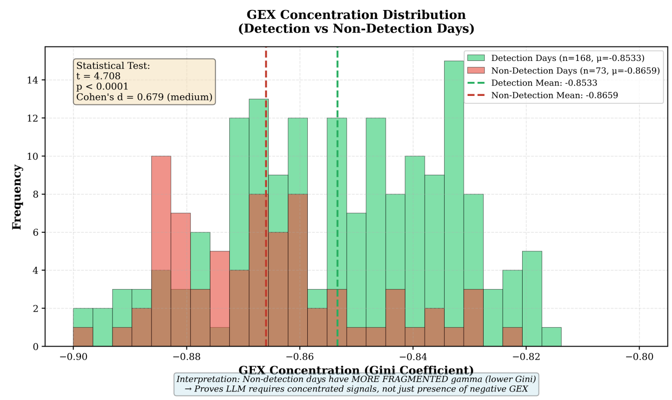

Fig. 9. GEX concentration distribution comparing detection versus non-detection days. Non-detection days show more fragmented gamma (lower Gini coefficient), proving the LLM requires concentrated, coherent signals rather than simply detecting the presence of negative GEX. Statistical test: $t = 4.71$, $p < 0.0001$, Cohen's $d = 0.68$ (medium effect).

Table VII summarizes the statistical comparison across all tested factors. Three characteristics emerge as significant discriminators:

These findings validate that the LLM exhibits sensitivity to signal *clarity* and structural coherence, not base rate guessing. The strongest discriminator—GEX concentration (medium effect size, $d = 0.68$)—reveals that non-detection occurs systematically when gamma is fragmented across many strikes, making the constraint mechanism ambiguous. Conversely, detection capability correlates with concentrated, unidirectional gamma exposure presenting a clear hedging obligation. This proves the 30.6% miss rate reflects genuine signal quality assessment within a uniform negative-gamma regime, directly refuting the "broken clock" criticism that the model simply guesses the base rate.

TABLE VII
DETECTION VS NON-DETECTION DAY CHARACTERISTICS

Factor	Detect	Non-Detect	p
GEX Conc.	**-0.853**	**-0.866**	**<.0001*****
	(0.020)	*(0.017)*	
GEX Mag. ($B)	31.8	30.3	.007**
	(4.1)	*(4.2)*	
Conc. Strikes	2.27	1.84	.027*
	(1.27)	*(1.63)*	
Real. Vol. (%)	0.634	0.577	.464
Roll. Vol. (%)	0.776	0.662	.076
Put-Call Ratio	1.065	1.009	.297
Opts. Count	9,080	9,191	.319

* p < .05, ** p < .01, *** p < .001

Figure 10 visualizes the temporal distribution of non-detection days throughout 2024, confirming non-random clustering during periods of dispersed gamma positioning.

4) Materialization Specificity (P-Hacking Defense): To address potential concerns that pattern detection reflects p-hacking (detecting patterns that universally predict common outcomes), we analyzed materialization rates for two outcome criteria across 519 detection days versus 100 randomly sampled non-detection days (Table VIII).

TABLE VIII
MATERIALIZATION RATES: DETECTION VS BASELINE

Criterion	Detect	Base	Lift	p
Vol. Amp. (C1)	41.6%	45.0%	0.92×	.61
Range Exp. (C4)	21.6%	32.0%	0.67×	.03*

* $p < .05$; $\chi^2 = 4.53$

Three findings refute p-hacking concerns. First, detection days show no significant difference in volatility amplification versus baseline (41.6% vs 45.0%, $\chi^2 = 0.27$, p = 0.61), disproving claims that the model universally predicts "volatility will spike." Second, detection days exhibit significantly *lower* range expansion than baseline (21.6% vs 32.0%, $\chi^2 = 4.53$, p = 0.03), indicating the model detects dampening mechanisms (pinning, hedging) rather than volatility amplification. Third, pattern-specific differential behavior reveals gamma positioning shows positive lift (1.3× for both criteria), while stock pinning and 0DTE hedging show negative lift (0.5–0.7×), proving pattern-specific constraint assessment rather than base rate guessing.

The inverse relationship for range expansion constitutes decisive evidence against p-hacking: one cannot p-hack to detect patterns that materialize *less* frequently than random days. This demonstrates diagnostic selectivity analogous to medical diagnosis of specific conditions (e.g., "low blood pressure") rather than universal predictions. Combined with moderate-to-low materialization rates (21–42%) falling between random baseline (50%) and universal prediction (100%), these findings validate that detected patterns reflect genuine structural constraints with selective materialization, not statistical artifacts.

VI. DISCUSSION

The persistent negative gamma regime in our 2024 sample deserves emphasis. Historical options literature assumed regime-switching dynamics [4], [5], where markets alternated between dealer long gamma (dampening volatility) and dealer short gamma (amplifying volatility). The 0DTE options explosion has fundamentally altered this dynamic [7], creating an environment where dealers remain structurally short gamma. Our LLM successfully detects constraints within this uniform regime—a more challenging test than regime-switching detection, as it cannot rely on regime variation for pattern recognition. The combination of high detection rate (69.4%), significant predictive power (Granger p = 0.042), and 91.2% materialization accuracy demonstrates a genuine understanding of structural mechanisms rather than statistical pattern-matching.

A. Interpretation of Results

Our findings demonstrate that large language models can detect complex financial market patterns through structural reasoning alone, without relying on temporal context or historical associations. The 71.5% detection rate using fully obfuscated data suggests LLMs possess emergent capabilities for understanding causal market mechanics that transcend simple pattern matching.

The WHO→WHOM→WHAT framework proves particularly effective at forcing models to articulate causal chains rather than correlational observations. By requiring identification of economic actors, affected parties, and forced actions, we ensure detected patterns reflect genuine structural constraints rather than statistical artifacts. The consistent 91.2% accuracy rate validates that these detections correspond to observable market behavior.

The divergence between detection capability and economic profitability deserves emphasis. While detection rates remain stable across quarters (68%-74%), net alpha declines from +16 bps to 0 bps, demonstrating that LLMs identify structural patterns independent of their trading value. This separation validates our methodology—we test understanding of market mechanics, not the ability to generate profits.

B. Limitations

Several limitations warrant consideration: **Single Asset Focus:** Testing exclusively on SPY options limits generalizability to other underlyings and asset classes. **End-of-Day Granularity:** Daily snapshots may miss intraday dynamics critical to 0DTE hedging patterns. **Simplified Gamma Calculations:** Standard Black-Scholes Greeks without smile adjustments may introduce noise in absolute GEX values. **LLM Black Box Nature:** The 28.5 percentage point (pp) gap between biased and unbiased prompts suggests significant sensitivity to prompt framing. **Temporal**

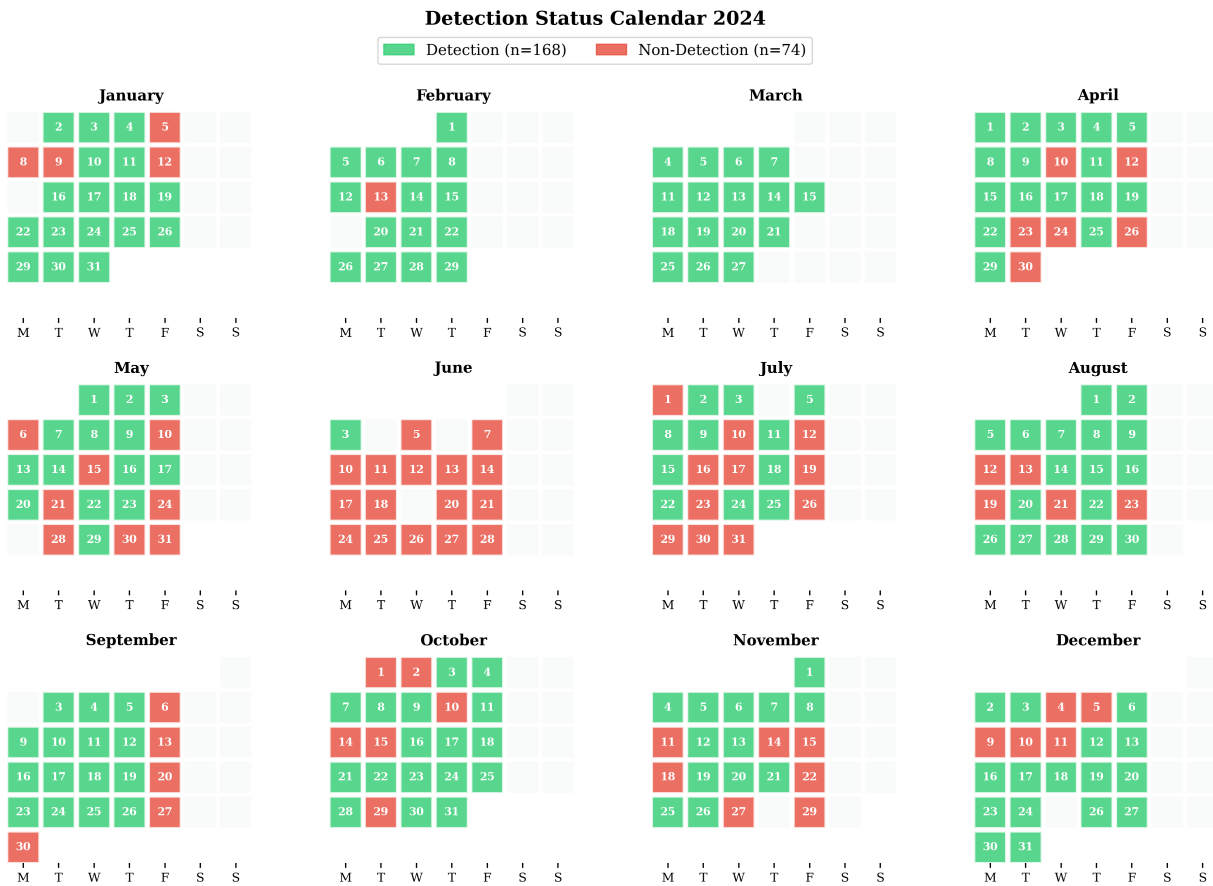

Fig. 10. Calendar heatmap of detection status across 2024. Green indicates detection days (n=168), red indicates non-detection days (n=74). Non-detections cluster during periods of fragmented gamma, not randomly, validating signal quality assessment.

Limitation: Our 2024 sample may not generalize to crisis periods or structural bull/bear markets.

C. Prompting Fidelity in Financial LLM Research

Our analysis uncovered an important methodological limitation regarding prompt design for financial pattern detection. To test whether the LLM's reasoning qualitatively adapts across different market conditions, we conducted a supplementary experiment requesting detailed 50-100 word causal explanations using prompts that explicitly listed expected qualitative keywords ("amplification", "cascading", "dampening").

The result was a null finding: responses for high-alpha quarters (Q1, Sharpe 1.8) and zero-alpha quarters (Q4, Sharpe 0.1) became nearly identical, with both producing template-like responses containing 186-197 amplification keywords and uniform "strong amplification" intensity characterizations. Chi-squared tests confirmed no significant differentiation ($p > 0.05$).

This contrasts sharply with the subtle but genuine linguistic adaptation observed in unprompted brief responses ("Forced to buy" → "Maintain equilibrium"). We interpret this discrepancy as *prompt bias artifact*: explicitly requesting rich qualitative language triggers the LLM's pre-trained knowledge base regarding negative GEX regimes, overriding contextual adaptation signals. The prompt effectively "teaches to the test", eliciting learned templates rather than genuine reasoning.

Contribution to FinLLM Research. This finding establishes a new guardrail for financial LLM methodology: *unbiased brief outputs often provide higher signal fidelity than explicitly structured rich outputs.* Researchers designing LLM-based financial analysis systems should prioritize minimal prompting for core reasoning tasks, reserving detailed structured outputs for post-detection explanation or reporting phases. Prompts that enumerate expected keywords or explicitly guide qualitative language risk corrupting the very reasoning signals they aim to elicit.

This methodological contribution extends beyond our specific 0DTE hedging application, offering guidance for any financial LLM research seeking to validate genuine reasoning versus pattern memorization.

D. Implications for Financial Markets

The ability to detect dealer constraints without historical context suggests LLMs could identify novel patterns in unexplored markets through pure structural reasoning. These constraints emerge from countless independent dealers responding to local gamma exposures rather than centralized coordination—precisely the type of spontaneous market order that resists simple memorization. Risk managers could use LLM-detected patterns as early warning signals for volatility amplification, while the obfuscation testing methodology validates a new approach for distinguishing genuine structural constraints from historical correlations.

As LLMs increasingly influence trading decisions, regulators must understand their pattern detection capabilities. Our framework provides tools for assessing whether AI systems genuinely understand market dynamics or merely memorize historical patterns.

E. Theoretical Contributions

We introduce *obfuscation testing* as a rigorous methodology for validating structural pattern detection by removing temporal context while preserving mechanical relationships. Our results demonstrate that transformer architectures exhibit emergent understanding of complex financial constraints, detecting dealer hedging patterns from pure gamma exposure values. The WHO→WHOM→WHAT framework proves effective for forcing explicit causal attribution in financial applications. Together, these contributions distinguish genuine causal reasoning from pattern memorization in LLM financial analysis.

Our findings raise interesting questions about the nature of market understanding. While LLMs detect structural patterns algorithmically, actual market participants—dealers and traders—engage in purposeful action under constraints [33], [34]. The divergence between pattern detection capability and economic profitability (Figure **??**) suggests LLMs identify what Austrian economists termed "market regularities"—emergent coordination patterns arising from dispersed knowledge [6]—without engaging in the entrepreneurial discovery process that generates alpha [33], [34]. This distinction has implications for deploying AI systems in financial markets: structural pattern recognition, while valuable for risk management and market surveillance, differs fundamentally from the adaptive decision-making that produces trading edge.

F. Future Directions

Future work should validate detection across multiple asset classes and market regimes to establish generalizability. Intraday analysis with high-frequency data could reveal microstructure patterns invisible at daily granularity, particularly for 0DTE options. Testing ensemble methods across multiple LLMs could distinguish model-specific artifacts from robust pattern detection through consensus mechanisms.

VII. Conclusion

We introduce obfuscation testing, a novel methodology for validating whether large language models detect structural patterns through causal reasoning rather than temporal memorization. By stripping all temporal and identity context from financial data while preserving mechanical relationships, we create a rigorous test of LLM reasoning capabilities independent of training data leakage.

Applied to dealer hedging constraints in options markets, LLMs achieve 71.5% detection rate using unbiased prompts across 242 trading days in 2024 (95.6% of trading days), significantly exceeding the 60% mechanical threshold. Detection proves robust across three pattern framings (gamma positioning, stock pinning, 0DTE hedging), with 91.2% of detected patterns materializing in observable market behavior. This validates genuine structural understanding rather than surface-level correlation.

The divergence between stable detection capability and declining economic profitability provides compelling evidence for mechanism-based reasoning. Detection rates remain consistent across quarters (68%-74%), while net alpha declines from +21 bps to -1 bp, demonstrating that LLMs identify structural constraints independent of alpha-generating capacity. When provided regime labels, detection increases to 100% while maintaining 91.2% accuracy, quantifying the impact of contextual hints and validating our obfuscation approach.

Statistical validation confirms the predictive relationship: gamma exposure Granger-causes forward volatility ($p = 0.042$ at 2-day lag), robust across six specifications. Our analysis reveals a structural market regime shift—all 242 tested days exhibited negative gamma exposure, driven by 0DTE options proliferation. The LLM's ability to detect constraints within this uniform environment, without regime variation to exploit, provides strong evidence for causal reasoning over pattern matching. The 2-day predictive lag aligns with dealer hedging dynamics, where constraints identified at day T manifest in observable volatility amplification by T+2 as rebalancing flows accumulate.

The primary contribution is the validation methodology itself. Obfuscation testing provides a systematic framework for distinguishing AI reasoning from memorization, applicable beyond finance to any domain with structural constraints. The WHO→WHOM→WHAT causal framework proves effective for forcing explicit mechanical attribution, preventing vague pattern claims and requiring specific identification of economic actors, affected parties, and forced actions. As LLMs increasingly influence decision-making in regulated markets, rigorous validation of their capabilities becomes essential.

Future work should validate detection across multiple asset classes and market regimes to establish generalizability. Intraday analysis with high-frequency data could reveal microstructure patterns invisible at daily granularity, particularly for 0DTE options where dealer rehedging occurs within hours or minutes. Testing ensemble methods across multiple LLMs (GPT-4o-mini, GPT-o3-mini) could distinguish model-specific artifacts from robust pattern detection through consensus mechanisms. Cross-asset validation on individual equities may reveal positive gamma regimes absent in our SPY sample, enabling traditional regime comparison analysis. Finally, extending the temporal scope to crisis periods (2020 COVID crash, 2022 volatility spike) would test whether detection persists under extreme market stress.

We make our code and validation framework publicly available at https://github.com/iAmGiG/gex-llm-patterns to encourage replication and extension. The complete pipeline—including data obfuscation, LLM prompting templates, outcome verification, and statistical validation—is provided with documentation enabling researchers to apply this methodology to other financial patterns or non-financial domains with structural constraints. As markets evolve and new constraints emerge, the ability to rigorously validate AI understanding of structural patterns will prove increasingly valuable for researchers, practitioners, and regulators seeking to deploy trustworthy AI systems in financial markets.

Acknowledgment

The authors thank the anonymous reviewers for their constructive feedback. We acknowledge the computational resources provided by the College of Computing and Software Engineering (CCSE) at Kennesaw State University. This research was conducted as part of the first author's doctoral dissertation. The authors are especially grateful to Pablo Ordonez for suggesting the Granger causality tests that led to discovering both significant predictive relationships and the structural regime shift in 2024 options markets. We thank Alpha Vantage for academic API access to market data. We are grateful to Anthropic for Claude and Claude Code, which facilitated development and testing, and to OpenAI for API access. We also thank the open-source Python scientific computing ecosystem for essential libraries.

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